

CSE 715: Computer Graphics

Chapter 11 — Color & Shading Models

Phong Model, RGB Interpolation, Monitor Color Matrix M , Point Left/Right Test, Color Gamut

University of Chittagong — 7th Semester B.Sc. (Engg.) Examination

Source: Schaum's Outline of Computer Graphics (Xiang & Plastock), Chapter 11 — this is the “Q7 cluster” (Section B) every year, occasionally spilling into Q6/Q8

Yield note (re-verified against all 5 raw scanned papers 2026-07-07): this chapter bundles **5 subtopics** that all live in the same exam-question cluster: RGB scanline interpolation (2021 2022 2023), Monitor color matrix M (2021 2023), Phong illumination model (2022 2023), Point left/right-of-line test (2020 2022), Color gamut (2022). **Every one of the 5 years has at least 2–3 of these**, so this is a real, guaranteed-to-recur Section-B question, not a stretch-goal. 2024 does not touch this cluster at all (its Q7/Q8 go to visible-surface/trapezoid-fill/ray-tracing instead). **Correction:** the Phong-model year attribution in `_TopicQuestionMap.md` previously said “2022 Q6d” — re-reading the raw 2022 scan confirms it is actually **Q7e**, not Q6d. Fixed in this pass.

Contents

1	Theoretical Framework	2
1.1	The Phong Illumination Model (§11.2)	2
1.2	Interpolative Shading Methods (§11.3)	2
1.3	RGB Linear Interpolation Along a Scanline	3
1.4	Monitor Color Transformation Matrix M (CIE XYZ)	3
1.5	Color Gamut of a Monitor	3
1.6	Point Left/Right-of-Line Test	4
2	Examination 2020 — Q7(a): Point Left/Right Test [2.25 marks]	4
3	Examination 2021 — Q7(a) & Q7(c)	4
4	Examination 2022 — Q7(a), Q7(d), Q7(e) & Q8(b)	5
5	Examination 2023 — Q6(d), Q7(b) & Q7(c)	6
6	Quick Summary	8

Theoretical Framework

The Phong Illumination Model (§11.2)

For a point Q on a surface, light source direction $\mathbf{L}$, surface normal $\mathbf{N}$, mirror-reflection direction $\mathbf{R}$, and viewer direction $\mathbf{V}$ (all unit vectors), the total reflected intensity is

$$I = I_a k_a + I_p (k_d \cos \theta + k_s \cos^n \varphi) = I_a k_a + I_p (k_d \mathbf{L} \cdot \mathbf{N} + k_s (\mathbf{R} \cdot \mathbf{V})^n)$$

where θ is the angle between $\mathbf{L}$ and $\mathbf{N}$ (angle of incidence), φ is the angle between $\mathbf{R}$ and $\mathbf{V}$, and $0 \leq k_a, k_d, k_s \leq 1$ are the surface's ambient/diffuse/specular reflection coefficients. $\mathbf{R} = 2(\mathbf{L} \cdot \mathbf{N})\mathbf{N} - \mathbf{L}$.

- **Ambient term** $I_a k_a$: directionless — a flat "fill" so unlit surfaces aren't pure black (accounts for indirect/bounced light).
- **Diffuse term** $I_p k_d \cos \theta$: models a *matte* surface scattering light equally in all directions; brightness depends only on the incidence angle θ (Lambert's law), **not** on viewer position — this is what gives an object its basic 3D "shape shading."
- **Specular term** $I_p k_s \cos^n \varphi$: models the *shiny highlight* on glossy surfaces, concentrated tightly around the mirror direction $\mathbf{R}$; **does** depend on viewer position $\mathbf{V}$. The shininess exponent n controls the highlight's tightness ($n = 1$: dull, $n = 100$: near-mirror).

For multiple light sources, the diffuse+specular term sums over all sources i : $I = I_a k_a + \sum_i I_{p_i} (k_d \mathbf{L}_i \cdot \mathbf{N} + k_s (\mathbf{R}_i \cdot \mathbf{V})^n)$.

Interpolative Shading Methods (§11.3)

Evaluating the Phong formula at every pixel is expensive, so real renderers evaluate it only at selected points and interpolate elsewhere:

- **Constant shading**: evaluate once per polygon (e.g. at its centroid), flat-fill the whole face. Fast, but false contours appear at polygon edges and specular highlights are often lost/distorted.
- **Gouraud shading**: evaluate the Phong formula at each *vertex*, then bilinearly interpolate the resulting *colors* across the polygon (via the two-edge-intersection method: interpolate along each edge to the scanline, then interpolate between those two points). Removes false contours even on a coarse mesh, but a fine mesh is still needed to capture sharp specular highlights.
- **Phong shading**: interpolate the *normal vectors* $\mathbf{N}$ (not the colors) across the polygon, then evaluate the full Phong formula at every pixel using the interpolated normal. More expensive, but handles specular highlights far better.

Vertex normals for a polygon mesh are obtained either from the underlying curved

surface (true geometric normal) or, absent one, by **averaging the normals of all polygons meeting at that vertex** (then renormalizing).

RGB Linear Interpolation Along a Scanline

Given a point P_1 with color/intensity I_1 on line y_1 and P_2 with color I_2 on line y_2 , the color at any line y' is found by linear interpolation, applied **component-wise** to (R, G, B) :

$$I' = I_1 + (I_2 - I_1) \cdot \frac{y' - y_1}{y_2 - y_1}$$

This is the same formula whether y' falls *between* y_1, y_2 (interpolation) or outside that range (extrapolation) — the fraction $\frac{y' - y_1}{y_2 - y_1}$ just falls outside $[0, 1]$ in the extrapolation case.

Monitor Color Transformation Matrix M (CIE XYZ)

Goal: given the monitor's white point (x_w, y_w, Y_w) (the XYZ chromaticity produced when $R = G = B = 1$) and the chromaticity coordinates $(x_r, y_r), (x_g, y_g), (x_b, y_b)$ of its three phosphors, find M such that $\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = M \begin{pmatrix} R \\ G \\ B \end{pmatrix}$. **Step 1 — white point to XYZ :** $X_w = \frac{x_w}{y_w} Y_w$, $Z_w = \frac{1 - x_w - y_w}{y_w} Y_w$. **Step 2 — auxiliary z 's:** $z_r = 1 - x_r - y_r$, $z_g = 1 - x_g - y_g$, $z_b = 1 - x_b - y_b$. **Step 3 — solve for C_r, C_g, C_b from**

$$\begin{pmatrix} X_w \\ Y_w \\ Z_w \end{pmatrix} = \begin{pmatrix} x_r & x_g & x_b \\ y_r & y_g & y_b \\ z_r & z_g & z_b \end{pmatrix} \begin{pmatrix} C_r \\ C_g \\ C_b \end{pmatrix}$$

Step 4 — assemble M :

$$M = \begin{pmatrix} x_r C_r & x_g C_g & x_b C_b \\ y_r C_r & y_g C_g & y_b C_b \\ z_r C_r & z_g C_g & z_b C_b \end{pmatrix}$$

Sanity check: each row of M must sum to X_w, Y_w, Z_w respectively (since $M \cdot (1, 1, 1)^T = (X_w, Y_w, Z_w)^T$ when $R = G = B = 1$).

Color Gamut of a Monitor

The **color gamut** of a monitor is the range of colors it can physically display. On the CIE chromaticity diagram, the chromaticity coordinates $(x_r, y_r), (x_g, y_g), (x_b, y_b)$ of the R/G/B phosphors form the three vertices of a **triangle**; any color *inside* that triangle can be produced as some non-negative combination of R, G, B , while more saturated colors *outside* the triangle cannot be reproduced exactly by that monitor. Different monitors (different phosphors) have different, generally non-identical gamut

triangles.

Point Left/Right-of-Line Test

Given a directed line segment from $P_1(x_1, y_1)$ to $P_2(x_2, y_2)$ and a test point $P(x, y)$, define

$$C = (x_2 - x_1)(y - y_1) - (y_2 - y_1)(x - x_1)$$

This is the z -component of the 2D cross product $\overrightarrow{P_1P_2} \times \overrightarrow{P_1P}$. In a standard right-handed xy system:

- $C > 0$: P lies to the **left** of the directed line $P_1 \rightarrow P_2$.
- $C < 0$: P lies to the **right** of the directed line $P_1 \rightarrow P_2$.
- $C = 0$: P lies **on** the line.

This single sign test is the building block for polygon-orientation checks, convex-polygon clipping (Sutherland-Hodgman "inside" tests), and back-face determination.

Examination 2020 — Q7(a): Point Left/Right Test [2.25 marks]

Question: Show the conditions to identify whether a point locates the right or left side of a line segment.

Derive $C = (x_2 - x_1)(y - y_1) - (y_2 - y_1)(x - x_1)$ as in §1.5 (cross-product z -component of $\overrightarrow{P_1P_2} \times \overrightarrow{P_1P}$), then state: $C > 0 \Rightarrow$ left, $C < 0 \Rightarrow$ right, $C = 0 \Rightarrow$ on the line.

Examination 2021 — Q7(a) & Q7(c)

Q7(a) — Monitor Color Matrix M [3 marks]

Question: Assume that a monitor produces the standard white D_{65} with $x_w = 0.313$, $y_w = 0.329$, $Y_w = 1.0$ when $R = G = B = 1$, and the chromaticity coordinates of its phosphors are $R(0.62, 0.34)$, $G(0.29, 0.59)$, $B(0.15, 0.06)$. Find the color transformation matrix M for the monitor.

$X_w = 0.313/0.329 = 0.951$, $Z_w = (1 - 0.313 - 0.329)/0.329 = 1.088$. Auxiliary z 's:

$z_r = 0.04, z_g = 0.12, z_b = 0.79$. Solving

$$\begin{pmatrix} 0.951 \\ 1.0 \\ 1.088 \end{pmatrix} = \begin{pmatrix} 0.62 & 0.29 & 0.15 \\ 0.34 & 0.59 & 0.06 \\ 0.04 & 0.12 & 0.79 \end{pmatrix} \begin{pmatrix} C_r \\ C_g \\ C_b \end{pmatrix} \Rightarrow C_r = 0.705, C_g = 1.170, C_b = 1.164$$

$$M = \begin{pmatrix} x_r C_r & x_g C_g & x_b C_b \\ y_r C_r & y_g C_g & y_b C_b \\ z_r C_r & z_g C_g & z_b C_b \end{pmatrix} = \begin{pmatrix} 0.437 & 0.339 & 0.175 \\ 0.240 & 0.690 & 0.070 \\ 0.028 & 0.140 & 0.920 \end{pmatrix}$$

Check: row sums = 0.951, 1.0, 1.088 ✓ matches X_w, Y_w, Z_w .

Q7(c) — RGB Scanline Interpolation [2 marks]

Question: If point P_1 on line 6 has an RGB color (1, 1, 0) and point P_2 on line 15 has an RGB color (0.6, 0.5, 0.2), what is the color for line 8?

$$\text{Fraction } t = \frac{8 - 6}{15 - 6} = \frac{2}{9} = 0.222.$$

$$R = 1 + (0.6 - 1)(0.222) = 0.911 \quad G = 1 + (0.5 - 1)(0.222) = 0.889 \quad B = 0 + (0.2 - 0)(0.222) = 0.044$$

Color at line 8 $\approx (0.911, 0.889, 0.044)$.

Examination 2022 — Q7(a), Q7(d), Q7(e) & Q8(b)

Q7(a) — Point Left/Right Test [1 mark]

Question: $C = (x_2 - x_1)(y - y_1) - (y_2 - y_1)(x - x_1)$. Refer to this expression, what are the conditions whether a point is left or right of a line segment?

$C > 0 \Rightarrow$ point is left of the directed segment $P_1 \rightarrow P_2$; $C < 0 \Rightarrow$ right; $C = 0 \Rightarrow$ on the line (see §1.5).

Q7(d) — RGB Scanline Interpolation [2.5 marks]

Question: If point p_1 on line 13 has an RGB color (0.4, 5, 0) and point p_2 on line 15 has an RGB color (2, 0.5, 7), what is the color for line 12?

Line 12 lies *outside* $[13, 15]$, so this is a linear **extrapolation** — same formula, fraction

$$t = \frac{12 - 13}{15 - 13} = -0.5.$$

$$R = 0.4 + (2 - 0.4)(-0.5) = -0.4 \quad G = 5 + (0.5 - 5)(-0.5) = 7.25 \quad B = 0 + (7 - 0)(-0.5) = -3.5$$

Color at line 12 = $(-0.4, 7.25, -3.5)$. (Values fall outside $[0, 1]$ because the given endpoint values themselves already exceed the normalized range — the question tests the interpolation *mechanism*, not physical color validity.)

Q7(e) — Significance of Diffuse and Specular Reflection [1.5 marks]

Question: Enumerate the significance of diffuse and specular reflection in the Phong illumination model.

- **Diffuse reflection** ($k_d \cos \theta$): models light scattered equally in all directions off a matte surface (Lambertian); intensity depends only on the incidence angle $\theta = \angle(\mathbf{L}, \mathbf{N})$, *not* on where the viewer is standing. This is what gives an object its basic, direction-independent 3D "form shading."
- **Specular reflection** ($k_s \cos^n \varphi$): models the shiny highlight seen on glossy surfaces, concentrated near the mirror-reflection direction $\mathbf{R}$; intensity depends on $\varphi = \angle(\mathbf{R}, \mathbf{V})$, so it *does* shift with viewer position. The exponent n (shininess) controls how tight/mirror-like the highlight looks.
- Together with the ambient term, they let a purely local (non-global) illumination model produce a visually convincing "shape + shiny highlight" appearance cheaply.

Q8(b) — Color Gamut of the Monitor [1 mark]

Question: Explain the color Gamut of the monitor.

See §1.6: the triangle in the CIE chromaticity diagram whose vertices are the monitor's R/G/B phosphor chromaticities; colors inside are displayable, colors outside (more saturated) are not.

Examination 2023 — Q6(d), Q7(b) & Q7(c)

Q6(d) — Explain the Phong Diagram [3 marks]

Question: In light of the Phong model, explain the following image (light source, point Q , vectors $\mathbf{L}$, $\mathbf{N}$, $\mathbf{R}$, $\mathbf{V}$, angles θ, θ, φ).

- $\mathbf{L}$: direction from surface point Q toward the light source.
- $\mathbf{N}$: surface normal at Q .
- $\mathbf{R}$: direction of perfect mirror reflection of $\mathbf{L}$ about $\mathbf{N}$ — by the law of reflection the angle between $\mathbf{L}$ and $\mathbf{N}$ equals the angle between $\mathbf{R}$ and $\mathbf{N}$ (both labeled θ in the figure).
- $\mathbf{V}$: direction from Q toward the viewpoint/eye.
- φ : the angle between $\mathbf{R}$ and $\mathbf{V}$ — the smaller φ is (viewer close to the mirror-reflection direction), the stronger the specular highlight, per $\cos^n \varphi$.

Together the figure is the geometric setup behind $I = I_a k_a + I_p (k_d \mathbf{L} \cdot \mathbf{N} + k_s (\mathbf{R} \cdot \mathbf{V})^n)$ from §1.1: the diffuse term depends only on θ (viewer-independent), the specular term depends on φ (viewer-dependent).

Q7(b) — Monitor Color Matrix M [3 marks]

Question: Assume that a monitor produces the standard white D_{65} with $x_w = 0.3$, $y_w = 0.2$, $Y_w = 1.0$ when $R = G = B = 1$, and the chromaticity coordinates of its phosphors are $R(0.62, 0.34)$, $G(0.29, 0.59)$, $B(0.15, 0.06)$. Find the color transformation matrix M .

$X_w = 0.3/0.2 = 1.5$, $Z_w = (1 - 0.3 - 0.2)/0.2 = 2.5$. Same phosphors as §3 (2021), but different white point, so solve fresh:

$$\begin{pmatrix} 1.5 \\ 1.0 \\ 2.5 \end{pmatrix} = \begin{pmatrix} 0.62 & 0.29 & 0.15 \\ 0.34 & 0.59 & 0.06 \\ 0.04 & 0.12 & 0.79 \end{pmatrix} \begin{pmatrix} C_r \\ C_g \\ C_b \end{pmatrix} \Rightarrow C_r \approx 1.427, C_g \approx 0.567, C_b \approx 3.007$$

$$M \approx \begin{pmatrix} 0.885 & 0.164 & 0.451 \\ 0.485 & 0.335 & 0.180 \\ 0.057 & 0.068 & 2.375 \end{pmatrix}$$

Check: row sums $\approx 1.500, 1.000, 2.500$ ✓ matches X_w, Y_w, Z_w .

Q7(c) — RGB Scanline Interpolation [2 marks]

Question: If point p_1 on line 3 has an RGB color $(5, 5, 0)$ and point p_2 on line 9 has an RGB color $(2, 0.5, 7)$, what is the color for line 5?

$$\text{Fraction } t = \frac{5-3}{9-3} = \frac{1}{3}.$$

$$R = 5 + (2-5)\frac{1}{3} = 4 \quad G = 5 + (0.5-5)\frac{1}{3} = 3.5 \quad B = 0 + (7-0)\frac{1}{3} = 2.333$$

Color at line 5 = (4, 3.5, 2.333).

Quick Summary

Subtopic	Years	Method
RGB scanline interpolation	2021, 2022, 2023	$I' = I_1 + (I_2 - I_1)\frac{y'-y_1}{y_2-y_1}$, component-wise; works for extrapolation too
Monitor color matrix M	2021, 2023	X_w, Z_w from white point $\rightarrow$ solve 3×3 for $C_r, C_g, C_b \rightarrow$ scale each phosphor column
Phong illumination model	2022, 2023	$I = I_a k_a + I_p(k_d \mathbf{L} \cdot \mathbf{N} + k_s(\mathbf{R} \cdot \mathbf{V})^n)$ — diffuse viewer-independent, specular viewer-dependent
Point left/right test	2020, 2022	$C = (x_2 - x_1)(y - y_1) - (y_2 - y_1)(x - x_1)$; $C > 0$ left, $C < 0$ right, $C = 0$ on line
Color gamut	2022	Triangle of phosphor chromaticities on CIE diagram = displayable color range

Exam-day takeaway: this whole cluster reduces to **five short, mechanical recipes** — none requires memorizing a proof, just the formula and the plug-in steps. The interpolation and matrix- M questions are pure arithmetic once you know the setup; Phong and color-gamut are one-paragraph explain answers built directly from §1.1 and §1.6.